

IIDA, Japan

WMO: 476370

Lat: 35.5233N

Lon: 137.8225E

Elev: 529

StdP: 95.13

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.5	-5.2	-14.8	1.1	1.9	-13.0	1.3	1.8	9.2	5.3	8.3	4.9	1.2	310	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.7	33.4	23.4	32.1	22.9	30.7	22.5	24.5	30.5	23.9	29.4	23.4	28.5	4.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	19.1	26.6	22.5	18.3	26.0	21.9	17.7	25.6	77.5	30.6	74.9	29.4	72.6	28.6	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.1	6.9	6.0	DB	-9.1	35.4	1.5	1.1	-10.2	36.1	-11.1	36.8	-11.9	37.3	-13.0	38.1	
			WB	-9.9	25.5	1.4	0.7	-10.9	26.1	-11.7	26.5	-12.5	26.9	-13.6	27.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	1.3	2.6	6.4	12.1	17.4	21.1	25.1	26.2	22.1	15.8	9.2	3.9
	DBStd	9.01	2.45	3.04	3.21	3.53	2.57	2.23	2.31	2.00	3.02	3.12	3.18	2.85
	HDD10.0	859	269	207	120	19	0	0	0	0	0	1	53	191
	HDD18.3	2398	527	440	371	188	49	4	0	6	91	274	448	
	CDD10.0	2194	0	1	8	82	228	333	469	501	362	181	29	1
	CDD18.3	691	0	0	0	1	19	86	211	243	118	13	0	0
	CDH23.3	5093	0	0	0	24	191	490	1642	2009	689	48	0	0
	CDH26.7	1797	0	0	0	2	29	94	647	827	194	3	0	0
Wind	WSAvg	2.1	2.1	2.2	2.5	2.6	2.4	2.2	2.2	2.1	1.8	1.7	1.7	2.1
Precipitation	PrecAvg	1535	60	76	121	124	142	176	203	141	200	142	91	60
	PrecMax	2069	142	163	239	256	329	357	460	362	351	372	209	142
	PrecMin	1002	4	21	58	20	46	55	38	36	74	14	19	2
	PrecStd	302	33	37	48	60	70	72	119	78	81	76	54	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	15.9	19.8	25.9	29.5	30.8	34.9	35.2	32.3	26.7	20.6	15.0
		MCWB	6.8	8.4	10.3	14.3	16.8	20.5	23.8	23.9	22.4	19.2	14.0	10.7
	2%	DB	9.7	12.4	17.1	23.3	27.2	29.0	33.4	33.7	30.6	24.6	18.3	12.5
		MCWB	4.9	6.6	8.6	12.9	16.3	19.7	23.4	23.5	21.8	17.7	12.7	8.1
	5%	DB	7.9	10.6	15.0	21.1	25.5	27.7	31.9	32.5	29.0	22.8	16.6	10.7
		MCWB	3.4	5.3	8.1	12.3	15.6	19.1	23.1	23.2	21.2	16.8	11.9	6.5
	10%	DB	6.3	8.5	13.0	19.0	23.8	26.3	30.4	31.2	27.3	21.2	14.9	9.2
		MCWB	2.1	3.9	7.5	11.6	15.0	18.8	22.7	23.0	20.6	16.2	11.0	5.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	10.9	13.1	17.8	20.4	22.9	25.6	25.4	23.7	20.9	16.5	12.2
		MCDB	10.1	13.0	14.9	20.2	24.2	27.3	31.8	32.0	28.6	23.9	18.0	13.6
	2%	WB	5.6	8.1	11.3	15.7	18.7	22.0	24.6	24.6	22.9	19.6	14.4	9.1
		MCDB	8.6	11.2	14.1	19.6	22.4	25.6	30.5	30.8	27.4	22.7	16.6	11.2
	5%	WB	3.9	6.0	9.8	14.2	17.6	21.2	23.9	24.1	22.4	18.4	12.9	7.3
		MCDB	7.1	9.1	13.6	18.5	22.2	24.6	29.5	29.8	26.6	21.5	15.5	9.8
	10%	WB	2.6	4.4	8.2	13.0	16.5	20.4	23.2	23.6	21.7	17.1	11.5	5.9
		MCDB	5.6	7.7	12.1	17.7	21.5	24.1	28.5	28.9	25.6	20.3	14.1	8.7
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.5	11.3	12.0	11.7	9.6	9.0	9.7	9.2	9.8	10.4	9.6
		MCDBR	11.7	13.4	14.7	15.2	14.9	11.9	11.7	11.6	11.0	11.3	12.3	11.8
		MCWBR	7.4	8.0	8.0	6.7	6.0	4.6	3.9	3.6	3.9	5.3	7.1	7.5
	5% WB	MCDBR	10.7	11.1	11.5	11.3	11.0	8.5	10.4	10.4	9.1	8.9	9.6	10.3
MCWBR		7.5	7.8	8.0	6.2	5.7	3.9	3.7	3.5	3.7	4.8	6.6	7.3	
Clear-Sky Solar Irradiance	taub	0.371	0.419	0.495	0.546	0.558	0.590	0.571	0.568	0.498	0.453	0.397	0.367	
	taud	2.138	2.009	1.864	1.806	1.831	1.824	1.933	1.930	2.081	2.123	2.196	2.205	
	Ebn at Noon	798	799	770	754	752	726	737	729	762	759	766	778	
	Edh at Noon	118	151	190	212	210	211	188	185	152	133	111	104	
All-Sky Solar Radiation	RadAvg	2.61	3.28	4.12	4.81	5.25	4.66	4.77	4.86	3.95	3.29	2.62	2.27	
	RadStd	0.17	0.22	0.28	0.46	0.57	0.39	0.67	0.45	0.35	0.33	0.21	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page